EE 354/CE 361/MATH 310 -

Introduction to Probability

and Statistics

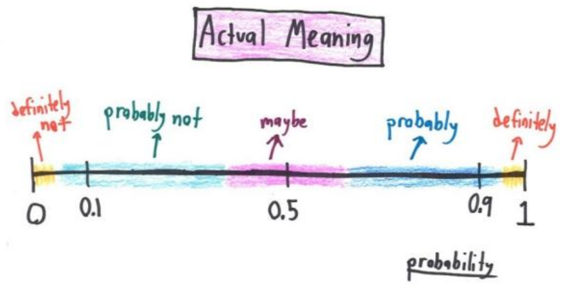
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 6 - Lecture No 11 – February 16, 2026



Announcements!

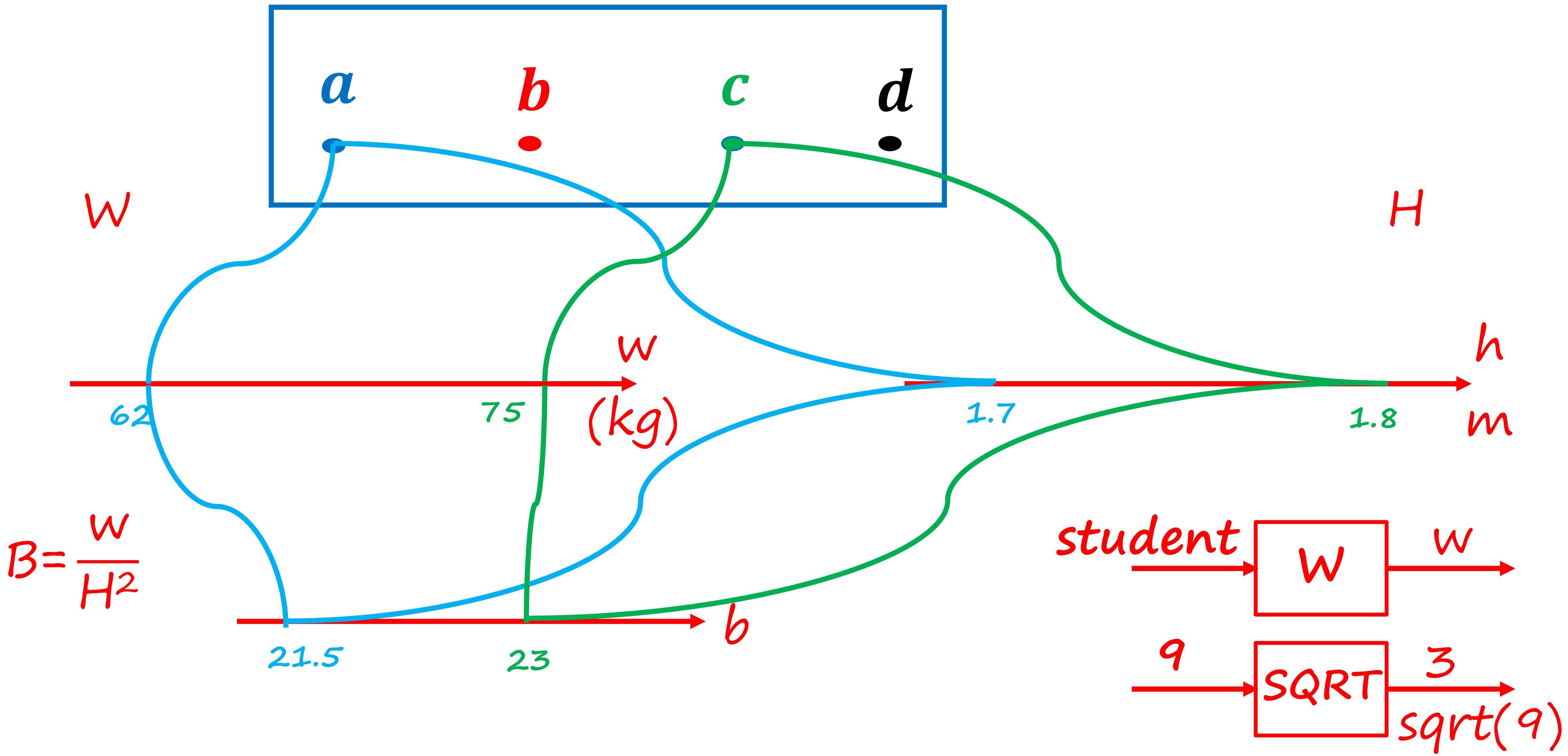
1. Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)
2. **Quiz No 04 (on DRV) is on February 25, 2026**
3. **Midterm Examination - March 3, 2026, from 03:25 PM to 04:55 PM at the Tariq Rafi Lecture Hall**
   - ✓ Two weeks remaining
   - ✓ Practice Problems uploaded
   - ✓ Midterm Exam for Fall 2025 uploaded

Agenda for today

1. Unit 4 – Discrete Random Variables (start)

Unit 4
Discrete Random Variables

Random Variables – The Idea



Random Variables – The formalism:

- A random variable (“r .v”) associates a value (a real number) to every possible outcome.
- **Mathematically:** A function from the sample space Ω to the real numbers.
- It can take discrete or continuous values

Notation: random variable X numerical value x

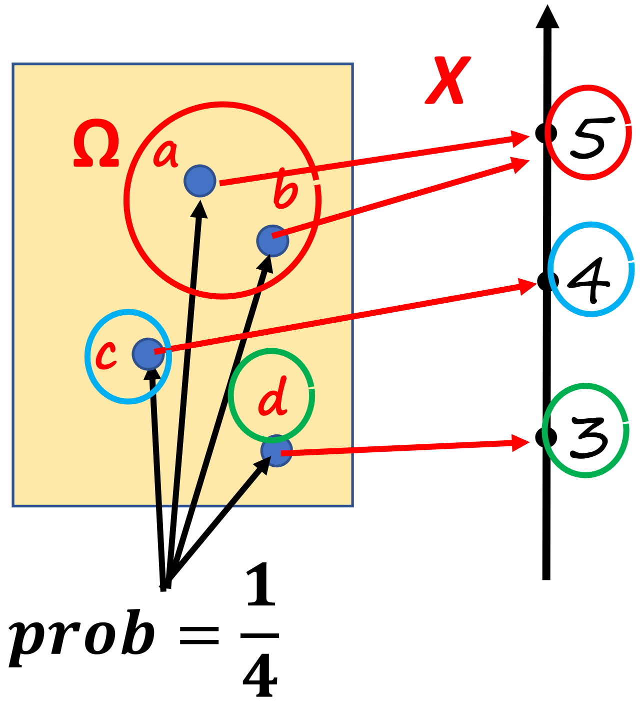
- We can have several random variables defined on the same sample space
- A function of one or several random variables is also a random variable
  - ❖ Meaning of $X + Y$: r.v takes value x+y.
    When X takes value x and Y takes value y

Probability mass function (PMF) of a discrete r.v. X

- It is the "probability law" or "probability distribution" of X
- If we fix some x, then "X = x" is an event

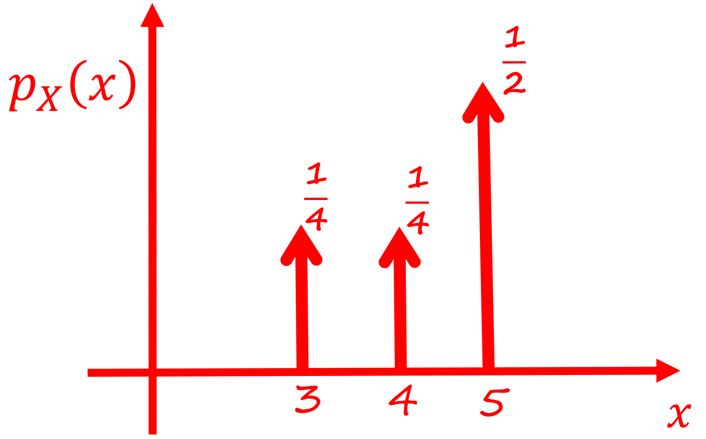
➡ $x=5, \quad X=5$ $\qquad \{ w : X(w) = 5 \} = \{ a, b \}$

➡ $P_X(5) = \frac{1}{2}$



$$p_X(x) = \mathbf{P}(X = x) = \mathbf{P}(\{ w \in \Omega \text{ s.t. } X(w) = x \})$$

- Properties: $\quad p_X(x) \geq 0 \qquad \sum_x p_X(x) = 1$



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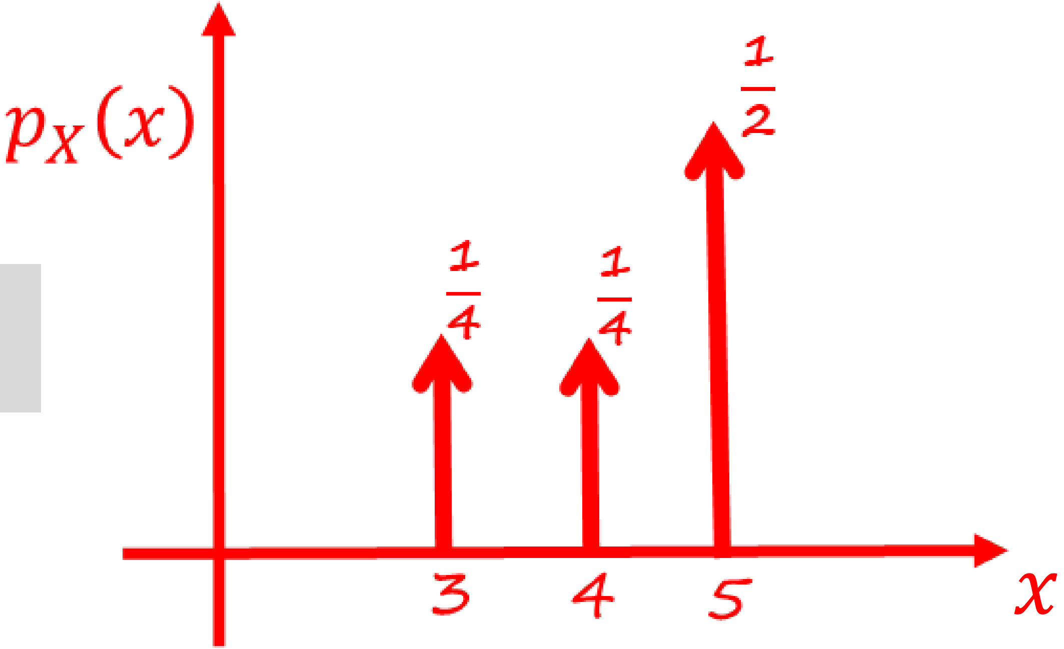


r.v X?

Sample space: discrete/finite example

- Two rolls of a tetrahedral die

Y = Second roll	X = 1	X = 2	X = 3	X = 4
4	5	6	7	8
3	4	5	6	7
2	3	4	5	6
1	2	3	4	5

X = First roll

- Let every possible outcome have probability 1/16

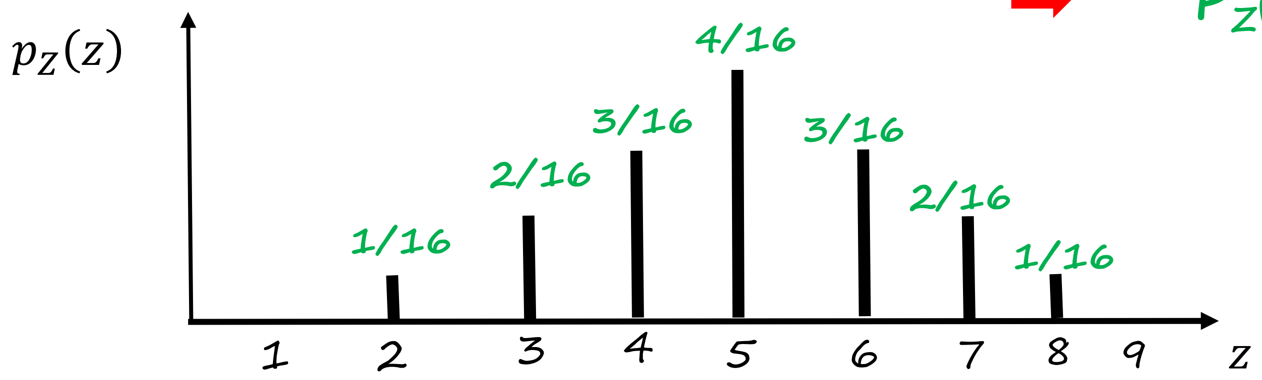
❖ $Z = X + Y$ Find $p_Z(z)$ for all z

- repeat for all z:
  - ✓ collect all possible outcomes for which Z is equal to z
  - ✓ add their probabilities

➡ $P_Z(2) = P(Z = 2) = 1/16$

➡ $P_Z(3) = P(Z = 3) = 2/16$

➡ $P_Z(4) = P(Z = 4) = 3/16$

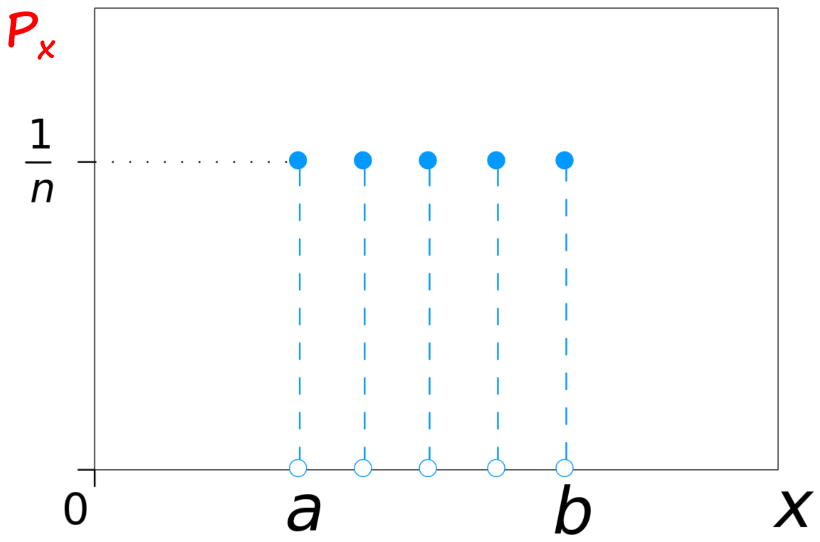


Two DRVs

Bernoulli Random Variable

parameters p

Discrete Uniform Random Variable



$1/n = 1/(b - a + 1)$

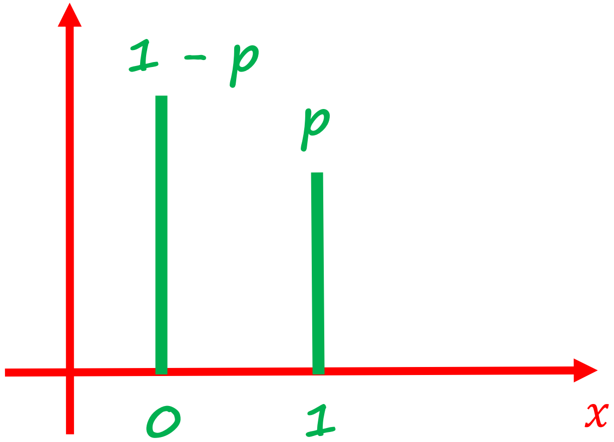
parameters a, b

The simplest Random variable; Bernoulli – parameters $p \in [0, 1]$

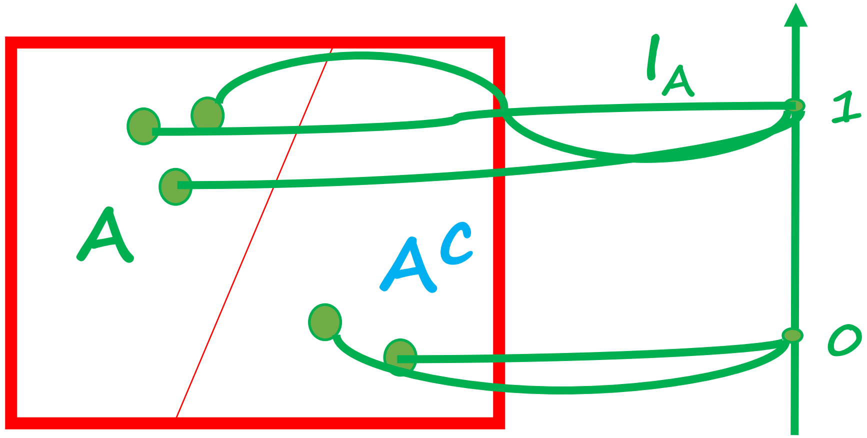
$$X = \begin{cases} 1, & \text{w.p.} \quad p \\ 0, & \text{w.p.} \quad 1-p \end{cases}$$

$P_X(0) = 1 - p$

$P_X(1) = p$



- Models a trial that results in success/failure, Heads/Tails, etc.
- Indicator r.v. of a event A: I_A = 1 iff A occurs

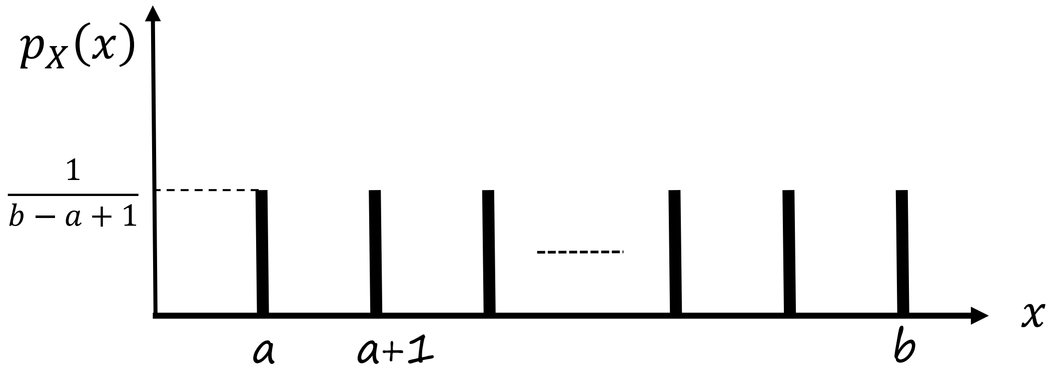


$$P_{IA}(1) = P(I_A = 1) = P(A)$$

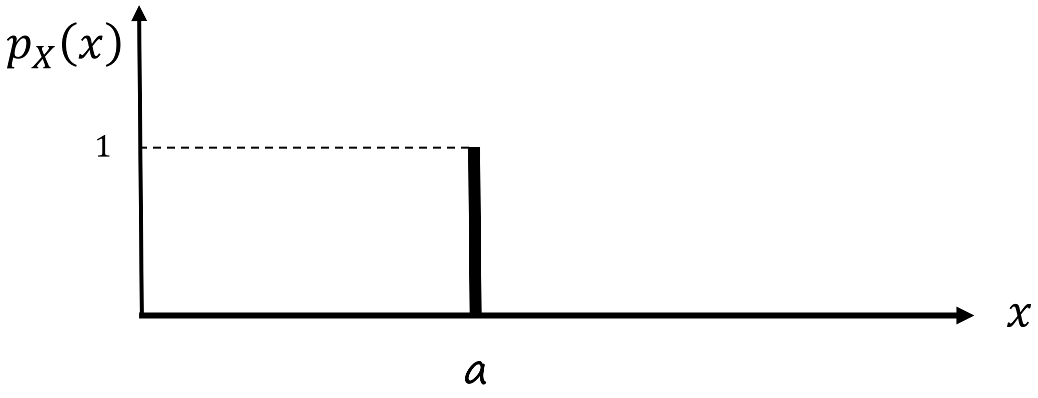
p

Discrete Uniform Random variable – parameters a, b

- **Parameters:** integers a, b
- **Experiment:** Pick one of a, a+1, … , b at random; all equally likely
- **Sample Space:** {a, a+1, …., b} $b - a + 1$ possible values
- **Random Variable *X*:** $\boldsymbol{X(\omega) = \omega}$ 11:52:29 $\{0, 1, \ldots, 59\}$
- **Model of:** complete ignorance



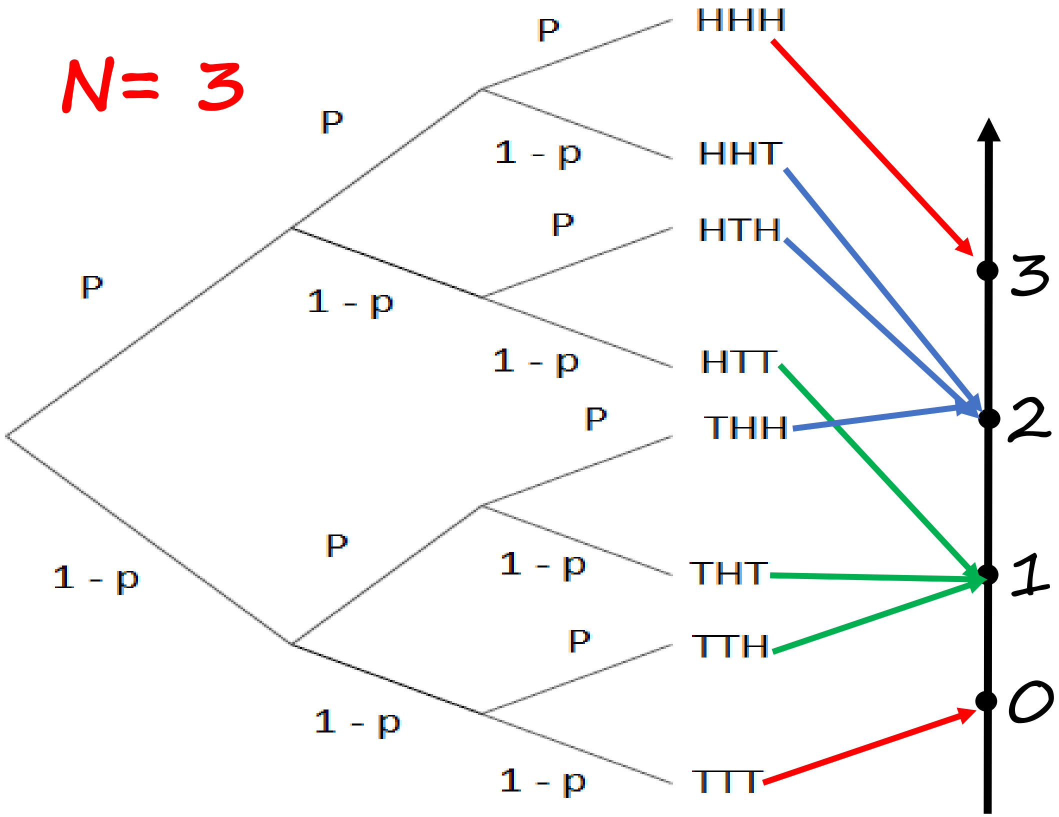
Special case: a = b



Binomial Random variable – parameters: positive integer *n*, $p \in [0,1]$

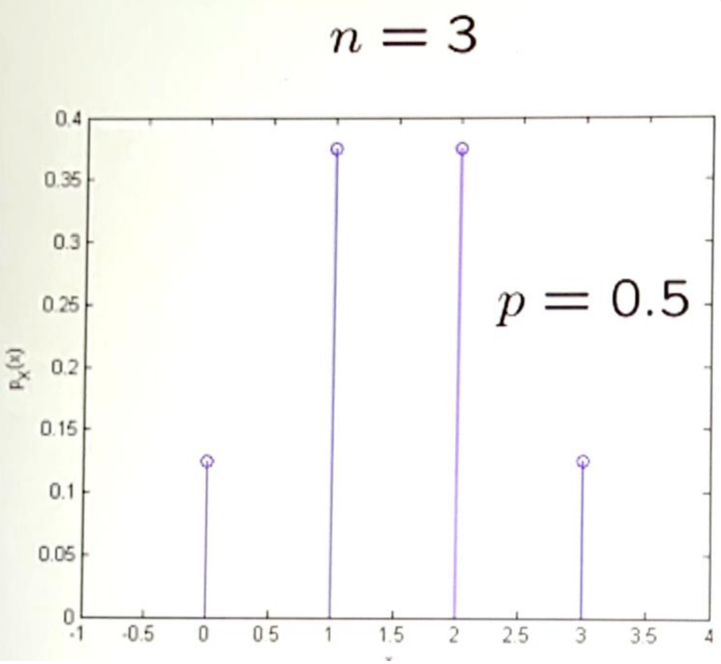
- **Experiment:** n independent tosses of a coin with **P**(Heads) = p
- **Sample Space:** Set of sequences of H and T, of length n
- **Random Variable *X*:** number of Heads observed
- **Model of:** number of successes in a given number of independent trials

N= 3



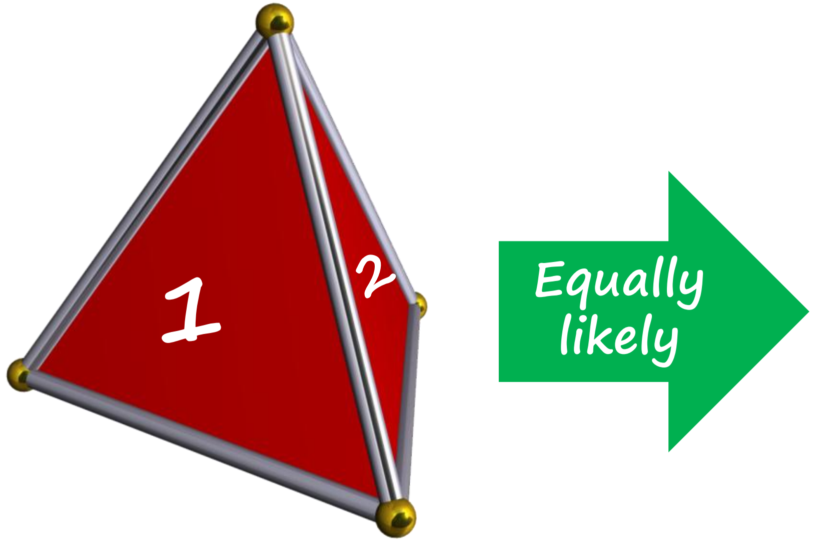
$$P_X(2) = P(X = 2)$$
$$= P(HHT) + P(HTH) + P(THH)$$
$$= 3p^2(1-p) = \binom{3}{2} p^2 (1-p)$$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$ *for k = 0, 1, …, n*



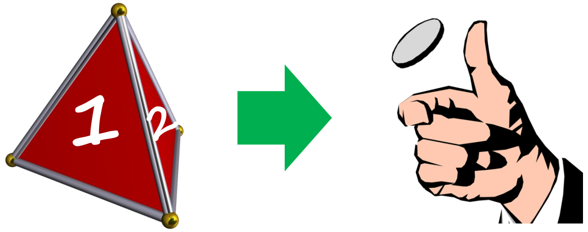
$n = 3$
$p = 0.5$
$p_X(x)$

Exercise No 01!



You throw a tetrahedral dice 12 times. R.V X is the number of times you get 3 in the 12 throws & $P_X(k)$ is the Probability of getting No 3, k-times in 12 throws

What is $P_X(5)$ =



parameters $p = 1/4$

$$p_X(5) = \binom{12}{5} \cdot \left(\frac{1}{4}\right)^5 \cdot \left(\frac{3}{4}\right)^7$$

Geometric Random variable – parameter: $p, \quad 0 < p \leq 1$

- **Experiment:** infinite many independent tosses of a coin with **P**(Heads) = p
- **Sample Space:** Set of infinite sequences of H and T
- **Random Variable *X*:** number of tosses until the first Heads
- **Model of:** waiting times; number of trials until a success

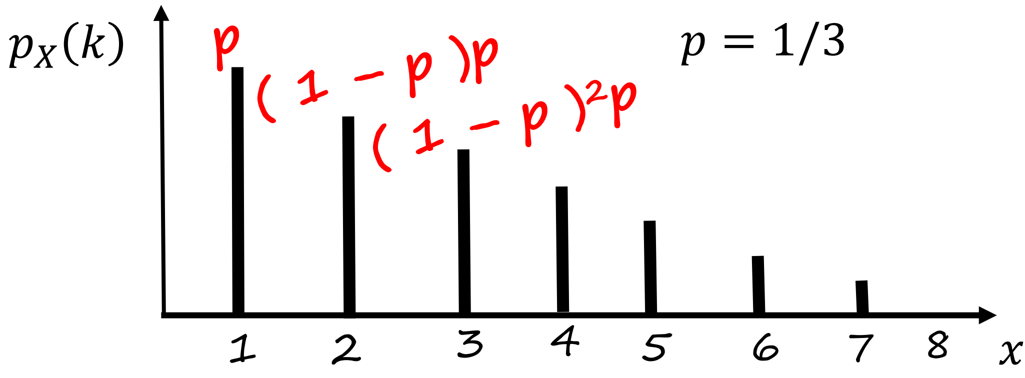
$\underbrace{TTTTH}_{X=5}HTT....$

➡ $p_X(x) = P(X = k) = P(T....T\,H) = (1-p)^{k-1}p \qquad k = 1, 2, 3, \ldots$

➡ P (no heads ever) $\leq P(T....T) = (1-p)^k \xrightarrow{k \to \infty} 0$

TTTTT....

"$X = \infty$" $= 0$



Geometric RV

Exercise No 02!

A graduating CS student of HU goes to career fair booths in technology sector – & his/her likelihood of receiving an off-campus interview invitation after a career fair booth visit depends on how well he/she did in EE 354 course. Specifically, an A in the course results in a probability $p_1 = 0.95$ of obtaining an invitation, whereas a C in EE 354 course results in a probability of $p_2 = 0.15$ of an invitation. Assuming that each student visits 5 booths during a typical career fair, find the probability that an A grade student in EE 354 course will not get an off-campus interview invitation. Similarly, find the probability that a C grade student in EE 354 course will get an invitation during a typical career fair

P(an A grade student does not get an invitation in 5 trials)= $\sum_{k=6}^{\infty}(1-p_1)^{k-1} \cdot p_1 = (1-p_1)^5$

P(C grade student gets an invitation in 5 trials)= $1-(1-p_2)^5$